

TupleProblems

le06 · 13jun26



```
In [1]:
import sys
sys.executable
```

```
Out [1]:
'/home/sivanagu/Desktop/PythonMCCP/.venv/bin/python'
```

```
In [4]:
t = tuple(list(map(int , input().split())))
print(t)
print(len(t))
```

```
Out [4]:
(10, 20, 30, 40, 50, 60)
6
```

```
In [15]:
t = sorted(tuple(list(map(int , input().split()))))
k = int(input())

print([t , k])
for i in range(k): print(t[i])
t = tuple(t[::-1])
print(type(t))
for i in range(k): print(t[i])
```

```
Out [15]:
[[10, 20, 30, 40, 50, 60], 2]
10
20
<class 'tuple'>
60
50
```

```
In [ ]:
l = list(map(int , input().split()))
l1 =[]
for x in l:
    l1.append((x , x**2 , x**3))
print(l1)
print(type(l1[0]))
```

```
Out [ ]:
[(1, 1, 1), (2, 4, 8), (3, 9, 27), (4, 16, 64)]
<class 'tuple'>
```

```
In [25]:
t = (tuple(list(map(int , input().split()))))

print (f"Using BUilt in :{sum(t)}")
s=0
for x in t:
    s += x
print(f"Using without Built in {s}")
```

```
Out [25]:
Using BUilt in :210
Using without Built in 210
```

```
In [26]:
t = (1 , 5 ,7 , 8 , 10)
t1 = tuple(i*j for i,j in zip(t,t[1:]))
```

```
print(t1)
```

```
Out [26]:  
(5, 35, 56, 80)
```

```
In [28]:
```

```
a = zip(t,t)  
print(list(a))
```

```
Out [28]:  
[(1, 1), (5, 5), (7, 7), (8, 8), (10, 10)]
```

```
In [3]:
```

```
def sort_tuples_by_digits(tuples_list):  
    def digits_in_tuple(t):  
        return sum(len(str(x)) for x in t)  
  
    return sorted(tuples_list, key=digits_in_tuple)
```

```
data = [(3, 4, 6, 723), (1, 2), (124, 324, 72)]  
print(sort_tuples_by_digits(data))
```

```
Out [3]:  
[(1, 2), (3, 4, 6, 723), (124, 324, 72)]
```

```
In [4]:
```

```
def tuples_all_divisible_by_k(tuples_list, k):  
    return [t for t in tuples_list if all(x % k == 0 for x in t)]
```

```
data = [(3, 4, 6, 723), (1, 2), (124, 324, 72)]  
print(tuples_all_divisible_by_k(data, 2))
```

```
Out [4]:  
[(124, 324, 72)]
```

```
In [5]:
```

```
def remove_tuples_with_length_k(tuples_list, k):  
    return [t for t in tuples_list if len(t) != k]
```

```
data = [(3, 4, 6, 723), (1, 2), (124, 324, 72)]  
print(remove_tuples_with_length_k(data, 2))
```

```
Out [5]:  
[(3, 4, 6, 723), (124, 324, 72)]
```

```
In [6]:
```

```
def sort_tuples_by_second(tuples_list):  
    return sorted(tuples_list, key=lambda t: t[1])
```

```
data = [(3, 10, 6), (1, 2), (124, 324, 72)]  
print(sort_tuples_by_second(data))
```

```
Out [6]:  
[(1, 2), (3, 10, 6), (124, 324, 72)]
```